SECTION 7.D
Solving Advanced Trigonometric Equations

Learning Objectives

- I can solve basic trigonometric equations visually for all six trig functions.
- I can solve linear trigonometric equations in radians involving secant, cosecant, and cotangent.
- I can solve advanced trigonometric equations in radians and verify solutions.
- I can attend to precision when writing general solutions.

Real-World Application

Trigonometric equations are used whenever motion repeats in cycles.

Examples include:

- The position of a Ferris wheel rider
- Tidal height changes
- Alternating current in electrical circuits
- Seasonal temperature models

Suppose the height h of a rider on a Ferris wheel is modeled by:

$$h(t) = 30 + 18\sin\left(\frac{\pi}{6}t\right)$$

If we want to know when the rider is exactly 29 feet above the ground, we solve:

$$30 + 18\sin\left(\frac{\pi}{6}t\right) = 29$$

This requires isolating the trig function and solving the equation within a time interval.

Introduction

In previous sections, we evaluated and rewrote trigonometric expressions.

Now we solve equations.

Solving trig equations requires:

- Isolating the trig function
- Finding reference angles
- Determining all solutions in an interval
- Writing general solutions when required

Precision matters.

Key Vocabulary

Principal Solution: Solutions within one cycle, typically $0 \leq \theta < 2\pi$

General Solution: All solutions written using appropriate period notation.

Period: Length of one complete cycle.

Core Strategy

- Step 1: Isolate the trig function.
- Step 2: Identify the reference angle.
- Step 3: Determine all solutions in the interval.
- Step 4: Write the general solution if required.

Examples

Example 1: Solve $\sin\theta = \frac{1}{2}$ for $0 \leq \theta < 2\pi$.

Example 2: Solve $2\cos\theta - 1 = 0$.

Example 3: Solve $\sec\theta = 2$.

Example 4: Solve $2\sin^2\theta - 1 = 0$.

Summary

- Solve algebra first.
- Use the unit circle for reference angles.
- Include all solutions in the interval.
- Use correct period when writing general solutions.

Common Mistakes

- Forgetting multiple quadrant solutions.
- Using the wrong period.
- Leaving answers in degrees.
- Stopping after one solution.

Practice Problems

Easy

1. Solve $\cos\theta = \frac{1}{2}$ for $0 \leq \theta < 2\pi$.

2. Solve $\sin\theta = -\frac{\sqrt{2}}{2}$.

3. Solve $\tan\theta = 1$.

4. Solve $\csc\theta = 2$.

Medium

5. Solve $2\sin\theta - \sqrt{3} = 0$.

6. Solve $\cos(2\theta) = 0$.

7. Solve $\cot\theta = -1$.

8. Solve $\sec\theta = -2$.

Harder

9. Solve $2\cos^2\theta - 1 = 0$.

10. Solve $\sin(3\theta) = \frac{\sqrt{3}}{2}$.

11. Solve $\tan^2\theta = 1$.

12. Solve $\cos\theta(2\sin\theta - 1) = 0$.

Challenge

13. Solve $2\sin^2\theta + 3\sin\theta - 2 = 0$.

14. Solve $\sec^2\theta - 4 = 0$.

15. Solve $\sin\theta + \cos\theta = 0$.

16. Solve $\tan(2\theta) = \sqrt{3}$.

Justify

17. Explain why tangent solutions use π instead of 2π.

18. Justify why solving $\sec\theta = 2$ requires rewriting in terms of cosine.

Solutions

Easy

1. $\frac{\pi}{3}, \frac{5\pi}{3}$

2. $\frac{5\pi}{4}, \frac{7\pi}{4}$

3. $\frac{\pi}{4}, \frac{5\pi}{4}$

4. $\frac{\pi}{6}, \frac{5\pi}{6}$

Medium

5. $\frac{\pi}{3}, \frac{5\pi}{3}$

6. $\frac{\pi}{4}, \frac{3\pi}{4}, \frac{5\pi}{4}, \frac{7\pi}{4}$

7. $\frac{3\pi}{4}, \frac{7\pi}{4}$

8. $\frac{2\pi}{3}, \frac{4\pi}{3}$

Harder

9. $\frac{\pi}{4}, \frac{3\pi}{4}, \frac{5\pi}{4}, \frac{7\pi}{4}$

10. Multiple solutions within interval

11. $\frac{\pi}{4}, \frac{3\pi}{4}, \frac{5\pi}{4}, \frac{7\pi}{4}$

12. Combine solutions from each factor

Challenge

13. Solve quadratic in sine

14. $\frac{\pi}{3}, \frac{5\pi}{3}$

15. $\frac{3\pi}{4}, \frac{7\pi}{4}$

16. Solve for 2θ, then divide by 2

Justify

17. Tangent has period π.

18. Secant is reciprocal of cosine.